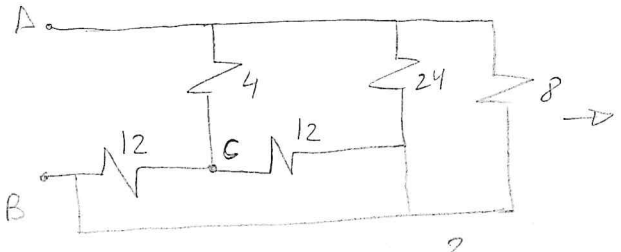
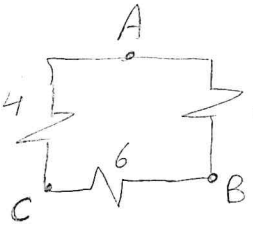
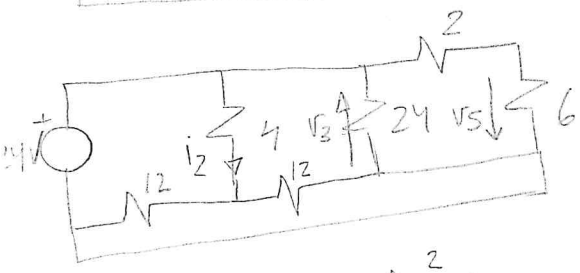
PARCIAL OCTUBRE 2026 (B)

1



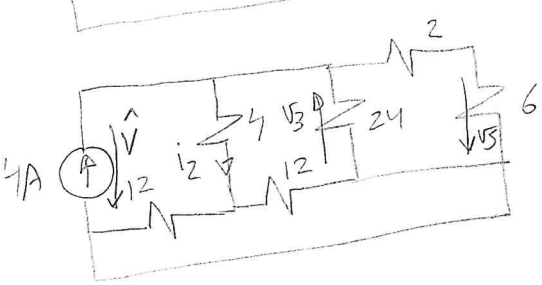


$R_{AB} = 10//6 = 3.75\Omega$



$V_3 = -24V \qquad V_5 = 24\,\frac{6}{6+2} = 18V$

$i_2 = \frac{24}{4+6} \Rightarrow i_2 = 2{,}4A. \qquad P_g = \frac{24^2}{3.75} = 153.6W$

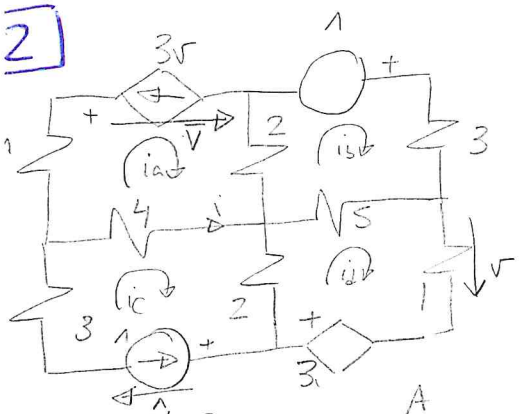


$\hat{V} = 4 \cdot 3{,}75 = 15V \qquad V_3 = -15V$

$V_5 = 15\,\frac{6}{6+2} = 11.25V \qquad i_2 = \frac{15}{6+4} = 1{,}5A.$

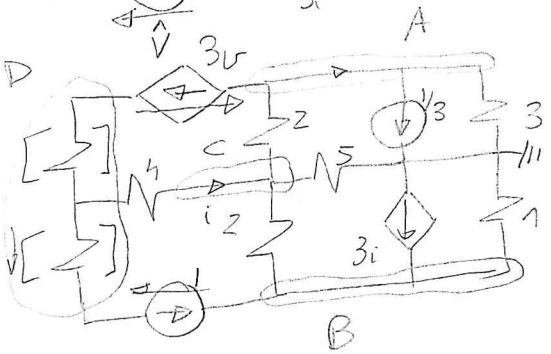
$P_g = 3.75 \cdot (4)^2 = 60W$

2



$$\begin{pmatrix} 7 & -2 & -4 & 0 \\ -2 & 10 & 0 & -5 \\ -4 & 0 & 9 & -2 \\ 0 & -5 & -2 & 8 \end{pmatrix} \begin{pmatrix} ia \\ ib \\ ic \\ id \end{pmatrix} = \begin{pmatrix} -\hat{V} \\ 1 \\ -\hat{V} \\ 3i \end{pmatrix}$$

$$\left.\begin{array}{l} 3V = -ia \\ 1 = -ic \\ i = ic - ia \\ V = id \end{array}\right\} \begin{array}{l} 8ec \\ 8incog \end{array}$$



$$\begin{pmatrix} 1/2+1/3 & 0 & -1/2 & 0 \\ 0 & 1+1/2 & -1/2 & 0 \\ -1/2 & -1/2 & 1/5+1/2+1/2+1/4 & -1/4 \\ 0 & 0 & -1/4 & 1/4 \end{pmatrix} \begin{pmatrix} V_A \\ V_B \\ V_C \\ V_D \end{pmatrix} = \begin{pmatrix} -1/3 - 3V \\ 1 + 3i \\ 0 \\ 3V - 1 \end{pmatrix}$$

$\left.\begin{array}{l} \\ \\ \\ \\ \end{array}\right\}$ 6ec 6incog

$V_B = 3i - V \qquad i = (V_D - V_C)/4$

Resuelvo por nudos

$3i = \frac{3}{4}(V_D - V_C) \qquad V = 3i - V_B \Rightarrow 3V = 9i - 3V_B \Rightarrow 3V = \frac{9}{4}(V_D - V_C) - 3V_B$

$$\begin{pmatrix} 1/2+1/3 & -3 & -1/2-9/4 & +9/4 \\ 0 & 1+1/2 & -1/2+3/4 & 0-3/4 \\ -1/2 & -1/2 & 1/5+1/2+1/2+1/4 & -1/4 \\ 0 & 0+3 & -1/4+9/4 & 1/4-9/4 \end{pmatrix} \begin{pmatrix} V_A \\ V_B \\ V_C \\ V_D \end{pmatrix} = \begin{pmatrix} -1/3 \\ 1 \\ 0 \\ -1 \end{pmatrix} \Rightarrow \begin{cases} V_A = -17/5V \\ V_B = 69V \\ V_C = 49V \\ V_D = 153V \end{cases}$$

$i = (V_D - V_C)/4 \Rightarrow i = +26A$

$V = 3i - V_B \Rightarrow V = +9V$

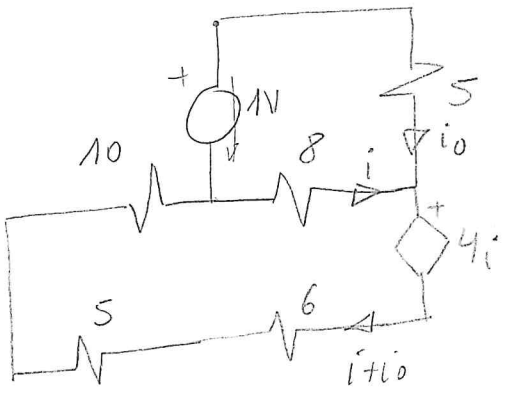
$P_{1V} = 1 \cdot (1/3 + V_A/3) \Rightarrow P_{1V} = -0.8W$

$P_{3V} = 3V \cdot (3V \cdot 1 + V_D - V_A) \Rightarrow P_{3V} = 4951,8W$

$P_{1A} = 1 \cdot (V_B - V_D + 1 \cdot 3) \Rightarrow P_{1A} = -81W$

$P_{3i} = 3i \cdot (3i - V_B/1) \Rightarrow P_{3i} = 702W$

3



$1 + 8i = 5i_0 \Rightarrow i_0 = \frac{1}{5}(1+8i)$

$4i + 21(i+i_0) + 8i = 0 \Rightarrow$

$\Rightarrow 33i + 21i_0 = 0 \Rightarrow$

$\Rightarrow 33i + \frac{21}{5}(1+8i) = 0 \Rightarrow$

$\Rightarrow 5 \cdot 33i + 21 + 21 \cdot 8i = 0 \Rightarrow i = \frac{-21}{5 \cdot 33 + 21 \cdot 8} \Rightarrow$

$\Rightarrow i = -0.06306A \quad i_0 = 0,099 \quad R_{th} = 1/i_0 = 10.09\Omega$ | $R^{max} = 10,09\Omega$ |